Minireview

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Matrix Metalloproteinases*

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The timely breakdown of extracellular matrix (ECM)[1] is essential for embryonic development, morphogenesis, reproduction, and tissue resorption and remodeling. The matrix metalloproteinases (MMPs), also called matrixins, are thought to play a central role in these processes. The expression of most matrixins is transcriptionally regulated by growth factors, hormones, cytokines, and cellular transformation (1, 2). The proteolytic activities of MMPs are precisely controlled during activation from their precursors and inhibition by endogenous inhibitors, α-macroglobulins, and tissue inhibitors of metalloproteinases (TIMPs). Table I lists currently known vertebrate matrixins. In addition, non-vertebrate members have been identified in sea urchins (3), *Caenorhabditis elegans* (4), soybean (5), and *Arabidopsis thaliana* (6). Most of these MMPs are the subject of individual chapters in the *Handbook of Proteolytic Enzymes* (7). This minireview focuses on recent progress in regulation of matrixin activities and their biological and pathological implications.

Domain Structure and Function

All matrixins are synthesized as prepro-enzymes and secreted as inactive pro-MMPs in most cases. The primary structures of 20 vertebrate matrixins comprise several domain motifs, as illustrated in Fig. 1; the domain composition for each MMP is listed in Table I.

The propeptide domain (about 80 amino acids) has a conserved unique PRCG(V/N)PD sequence. The Cys within this sequence (the "cysteine switch") ligates the catalytic zinc to maintain the latency of pro-MMPs (8, 9). This sequence is missing in MMP-23 (10). Stromelysin 3 (MMP-11), MT-MMPs, *Xenopus* MMP, and MMP-23 have a proprotein processing sequence R*X*(K/R)R at the C-terminal end of the propeptide, and MMP-11 (11) and MMP-14 (12) were shown to be activated intracellularly by furin.

The catalytic domain (about 170 amino acids) contains a zinc binding motif H*EXXHXXGXX*H and a conserved methionine, which forms a unique "Met-turn" structure (13). This domain consists of a five-stranded β-sheet, three α-helices, and bridging loops (14). These backbone structures including the "Met-turn" are similar to those of the members from other metalloproteinase families, *i.e.* astacins, reprolysins (ADAMs), and serralysins; these four families constitute the "metzincins" (13). The catalytic domains of matrixins have an additional structural zinc ion and 2–3 calcium ions, which are required for the stability and the expression of enzymic activity. MMP-2 and MMP-9 have three repeats of fibronectin-type II domain inserted in the catalytic domain. These repeats interact with collagens and gelatins (15, 16).

The C-terminal hemopexin-like domain (about 210 amino acids) has an ellipsoidal disk shape with a four bladed β-propeller structure; each blade consists of four antiparallel β-strands and an α-helix (17). The hemopexin domain is an absolute requirement for collagenases to cleave triple helical interstitial collagens (18), although the catalytic domains alone retain proteolytic activity toward other substrates (19). The hemopexin domain of MMP-2 is also required for the cell surface activation of pro-MMP-2 by MT1-MMP (20, 21). The function of the proline-rich linker peptide that connects the catalytic and the hemopexin domains is not known, although its interaction with triple helical collagen is hypothesized based on molecular modeling (22). MMP-23 has cysteine-rich, proline-rich, and IL-1 receptor-like regions instead of the hemopexin domain (10). A transmembrane domain is found in the MT-MMPs, which anchors those enzymes to the cell surface.

Regulation of Matrixin Gene Expression

One of the striking features of the matrixins is that many of those genes are "inducible." The effectors include growth factors, cytokines, chemical agents (*e.g.* phorbol esters, actin stress fiber-disrupting drugs), physical stress, and oncogenic cellular transformation, etc., and the enhanced MMP gene expression may be down-regulated by suppressive factors (*e.g.* transforming growth factor β, retinoic acids, glucocorticoids) (1). Induction and suppression through the promoter regions of matrixin genes have recently been reviewed by Fini *et al*. (2).

Recent studies emphasize not only soluble factors but also cell-matrix and cell-cell interactions as keys in gene expression of matrixins. Examples are: induction of MMP-1, -2, and -3 in fibroblasts by EMMPRIN (M6 antigen or basigin), a member of the immunoglobulin family expressed on tumor cell surface (23); induction of MMP-9 in T lymphoma cells through leukocyte function-associated antigen-1 (LFA-1)-intercellular adhesion molecule-1 (ICAM-1)-mediated cell adhesion (24); induction of MMP-2 in T cells through very late antigen 4 (VLA-4)-vascular cell adhesion molecule-1 (VCAM-1)-mediated adhesion to endothelial cells (25); MMP-9 expression in monocytes by the activated T cells through gp39-CD40 interaction (26); and $\alpha_5\beta_1$ integrin-fibronectin interaction for MMP-9 expression during macrophage differentiation (27). Endothelial cells (28, 29), fibroblasts (30), and neoplastic cells (31) cultured in type I collagen gel express MT1-MMP, which appears to be mediated by $\alpha_2\beta_1$ integrin (30).

Certain signaling pathways lead to expression of a particular MMP gene. Addition of soluble antibody to $\alpha_5\beta_1$ integrin causes disruption of the actin cytoskeleton and an increased expression of MMP-1 in rabbit synovial fibroblasts (32). This is because of activation of the GTP-binding protein Rac1, which generated reactive oxygen species and induced activation of NF-κB (33). This leads to induction of IL-1α, an autocrine inducer of MMP-1 expression. This pathway is distinct from integrin-dependent spreading, which leads to MMP-1 expression mediated through AP-1 and PEA3-binding sites (34) but does not require the Rho-family GTPase pathway (33).

Inflammatory cytokines, TNF-α and IL-1, trigger the ceramide signaling pathways (35). The ceramide-dependent expression of MMP-1 in human skin fibroblast is mediated by three distinct MAP kinase pathways, *i.e.* ERK1/2, stress-activated protein kinase (SAPK)/JNK, and p38 (36). The maximal induction of MMP-1 gene expression in human astrocytes by oncostatin M is Raf1-dependent and requires cross-talk between the Janus kinase (JAK)-signal transducer and activator of transcription (STAT) and MAP kinase pathways (37).

Ultraviolet B irradiation up-regulates MMP-1, MMP-3, and MMP-9 expression in human dermal fibroblasts (38). Brenneisen *et al.* (39) reported that this was mediated by the activation of

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[1] The abbreviations used are: ECM, extracellular matrix; AP-1, activator protein 1; ATF-2, activating transcription factor 2; ERK1/2, extracellular signal-regulated kinase 1/2; IL-1, interleukin 1; JNK, Jun N-terminal kinase; MAP, mitogen-activated protein; MMP, matrix metalloproteinase; MT-MMP, membrane-type matrix metalloproteinase; N-TIMP, N-terminal domain of tissue inhibitor of metalloproteinases; TIMP, tissue inhibitor of metalloproteinases; TNF-α, tumor necrosis factor α; uPA, urokinase-type plasminogen activator.

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TABLE I
Vertebrate members of the matrixin family
See Fig. 1 for domain composition.

Protein	MMP	Domain composition
Collagenase 1	MMP-1	B
Gelatinase A	MMP-2	C
Stromelysin 1	MMP-3	B
Matrilysin	MMP-7	A
Collagenase 2	MMP-8	B
Gelatinase B	MMP-9	D
Stromelysin 2	MMP-10	B
Stromelysin 3	MMP-11	E
Macrophage elastase	MMP-12	B
Collagenase 3	MMP-13	B
MT1-MMP	MMP-14	F
MT2-MMP	MMP-15	F
MT3-MMP	MMP-16	F
MT4-MMP	MMP-17	F
Collagenase 4 (*Xenopus*)	MMP-18	B
(No trivial name)	MMP-19	B
Enamelysin	MMP-20	B
XMMP (*Xenopus*)	MMP-21	G
CMMP (chicken)	MMP-22	B
(No trivial name)	MMP-23	H

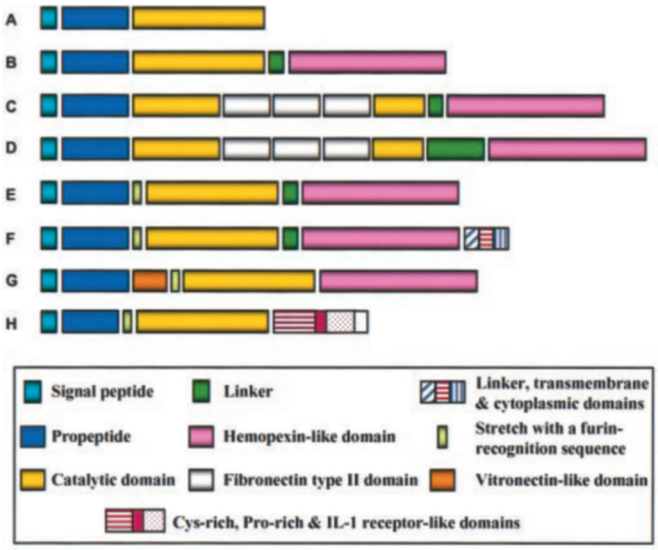


FIG. 1. **Domain arrangements of vertebrate matrixins.**

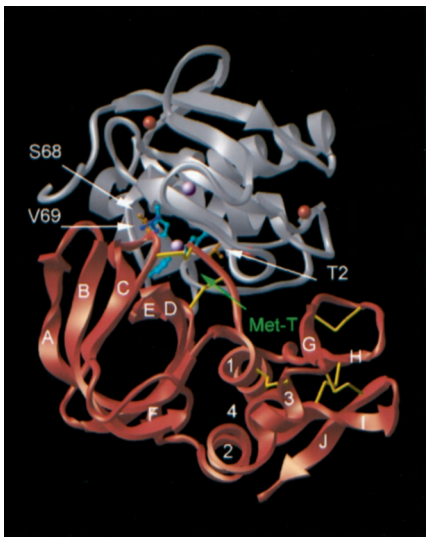


FIG. 2. **Ribbon diagram of the complex of TIMP-1 and the catalytic domain of MMP-3.** The image was prepared from the Brookhaven Protein Data Bank entry (1UEA) using RIBBONS (91). MMP-3 is shown in *silver* and TIMP-1 in *red*. The catalytic and structural zinc ions and the three calcium ions in MMP-3 are shown as *purple* and *red spheres*, respectively. Three histidines that ligate the catalytic zinc ion are shown in *blue*. Strands and helices in TIMP-1 are labeled as *A–J* and *1–4*, respectively. Disulfide bonds in TIMP-1 are in *yellow*. The positions of "Met-turn" (*Met-T*) in MMP-3, and Thr^2 (*T2*), Ser^{68} (*S68*), and Val^{69} (*V69*) in TIMP-1 are indicated.

stress-activated protein kinase JNK-2 through the generation of reactive oxygen species and lipid peroxidation by the ion-driven Fenton reaction. The exposure of human skin to ultraviolet B *in vivo* activates epidermal growth factor receptor and the GTP-binding regulatory protein Ras, and stimulates ERK, JNK, and p38 MAP kinase pathways, which phosphorylate and activate c-Jun and ATF-2. c-Jun and ATF-2 heterodimers in turn bind to the c-Jun promoter and up-regulate c-Jun, which together with c-Fos increased levels of AP-1 required for MMP transcription (40).

Activation of Pro-matrixins: Cell Surface Activation of Pro-MMP2

Apart from a few members activated by furin (see above), most are secreted from the cell as inactive zymogens. Secreted pro-matrixins are activated *in vitro* by proteinases and by non-proteolytic agents such as SH-reactive agents, mercurial compounds, reactive oxygen, and denaturants. In all cases activation requires the disruption of the Cys-Zn^{2+} (cysteine switch) interaction, and the removal of the propeptide proceeds often in a stepwise manner (41). *In vivo*, most pro-matrixins are likely to be activated by tissue or plasma proteinases or opportunistic bacterial proteinases. Using transgenic mice deficient in uPA, Carmeliet *et al.* (42) have suggested that the uPA/plasmin system is a pathophysiologically significant activator of pro-matrixins. The activation of pro-MMP-2, on the other hand, is thought to take place primarily on the cell surface.

In 1994, Sato *et al.* (43) cloned the first membrane-type MMP (MT1-MMP) and demonstrated it to be an activator of pro-MMP-2. Recent studies propose that this activation process requires both active MT1-MMP and the TIMP-2-bound MT1-MMP (21, 44, 45).The TIMP-2 in the latter complex binds, through its C-terminal domain, to the hemopexin domain of pro-MMP-2, which is assumed to localize the zymogen close to the active MT1-MMP (44). Although the major pro-MMP-2 activation pathway appears to be through MT-MMPs, Mazzieri *et al.* (46) reported that surface-bound pro-MMP-2 also can be activated by the cell surface-associated uPA/plasmin system, whereas soluble plasmin degrades it.

The expression of MT1-MMP and activation of pro-MMP-2 in tumor cells occur in specialized membrane extensions, invadopodia (47). This focal location of MT1-MMP is controlled by the transmembrane and cytoplasmic domains of MT1-MMP. Hiraoka *et al.* (48) reported that specific fibrinolysis by the plasma membrane-anchored MT1-MMP in endothelial cells is critical for neovascularization. MMP-2 may also be localized on the cell surface of angiogenic endothelial cells and melanoma cells by binding to $\alpha_v\beta_3$ integrin (49). This interaction is inhibited by a RGD peptide and an excess C-terminal hemopexin domain of MMP-2, which prevents invasive behavior of new blood vessels (50). In contrast, binding of pro-MMP-2 to mouse melanoma cells was not inhibited by the RGDS peptide (51), suggesting that the modes of MMP-2 binding to the cell surface may vary. Those studies, nonetheless, emphasize that membrane-bound proteinases such as MT1-MMP and MMP-2 play an important role in angiogenesis and tumor cell invasion.

Inhibition Mechanism of TIMPs

TIMPs (21–30 kDa) are the major endogenous regulators of MMP activities in the tissue, and four homologous TIMPs (TIMPs-1 to -4) have been identified to date (52). The crystal structure of the complex formed between TIMP-1 and the catalytic domain of MMP-3 was determined by Gomis-Rüth *et al.* (53). The critical residues involved in MMP inhibition are located around the disulfide bond between Cys^1 and Cys^{70} (Fig. 2). The N-terminal α-amino and carbonyl groups of Cys^1 bidentately coordinate the catalytic Zn^{2+}. The N-terminal segment Cys^1-Thr-Cys-Val-Pro^5 binds to the active site cleft subsite S1 to S4′ of the enzyme like the substrate, whereas Ser^{68} and Val^{69} in the connecting loop between

strand C and strand D are arranged in an orientation nearly opposite to that of the P3-P2 segment of a substrate. The side chain of Thr^2 extends into the large S1′ specificity pocket of MMP-3. TIMP-2 binds to matrixins in a similar manner but has a longer strand A-strand B hairpin loop (54). The high resolution NMR structure of N-TIMP-2 (55) and the crystal structure of free TIMP-2 (56) were also reported. Proteinase binding causes an internal rotation between the N-terminal and the C-terminal domains of ~13° and other local conformational changes, especially in the AB β-hairpin loop in TIMP-2 (56). MMP-3 contact sites in N-TIMP-2 (55) and N-TIMP-1 contact sites in MMP-3 (57) were also mapped by NMR.

TIMPs Are Multifunctional Proteins

It is widely appreciated that TIMPs inhibit cell invasion *in vitro*, tumorigenesis, metastasis *in vivo*, and angiogenesis (52). TIMPs exhibit additional biological functions. TIMP-1 and TIMP-2 have mitogenic activities on a number of cell types, whereas overexpression of these inhibitors reduces tumor cell growth (52), and TIMP-2, but not TIMP-1, inhibits basic fibroblast growth factor-induced human endothelial cell growth (58). These biological activities of TIMPs are independent of MMP-inhibitory activities (59, 60). The TIMP-1-procathepsin L complex was reported to stimulate steroidogenesis of rat testis and ovary *in vitro* (61), but TIMP-1-deficient mice gave little evidence for regulation of steroidogenesis *in vivo* (62). TIMP-1 also stimulates fibroblasts to produce MMP-1 (63). Recent studies of Zhao *et al.* (64) demonstrated TIMP-1 accumulates in the nuclei of human fibroblasts in a cell cycle-dependent manner (maximal at the S phase), suggestive of its participation in cell growth.

More recently, TIMP-3 was shown to induce apoptosis of human colon carcinoma cells (65) and melanoma cells (66). This is probably because of stabilization of TNF-α receptors (65); TIMP-3 inhibits TNF-α-converting enzyme (67) and IL-6 receptor shedding (68). Overexpression of TIMP-3 in rat vascular smooth muscle cells also promoted apoptosis, but this effect was independent of MMP inhibition (69). Similar independence is found when TIMP-1 suppresses B cell apoptosis induced by CD95-dependent and -independent (cold shock, serum deprivation, and γ-radiation) pathways (70).

TIMP-3 binds to extracellular matrix (71) through the C-terminal domain of the inhibitor (72). A single mutation in the C-terminal domain (S156C, G166C, G167C, Y168C, or S181C) is associated with Sorsby's fundus dystrophy, an autosomal dominant macular disorder that causes irreversible loss of vision with onset in the third and fourth decade (73). TIMPs are therefore important regulators not only in matrix turnover but also in cellular activities.

Biological and Pathological Roles of Matrixins

Matrixins participate in many normal biological processes (*e.g.* embryonic development, blastocyst implantation, organ morphogenesis, nerve growth, ovulation, cervical dilatation, postpartum uterine involution, endometrial cycling, hair follicle cycling, bone remodeling, wound healing, angiogenesis, apoptosis, etc.) and pathological processes (*e.g.* arthritis, cancer, cardiovascular disease, nephritis, neurological disease, breakdown of blood brain barrier, periodontal disease, skin ulceration, gastric ulcer, corneal ulceration, liver fibrosis, emphysema, fibrotic lung disease, etc.) (74). Although the main function of matrixins is removal of ECM during tissue resorption and progression of many diseases, it is notable that MMPs also alter biological functions of ECM macromolecules by specific proteolysis. For example, MMP-2 released by growth cones promotes neurite outgrowth by inactivating neurite-inhibitory chondroitin sulfate proteoglycans, thereby unmasking the neurite-promoting activity of laminin (75). Specific cleavage of the Ala^{586}-Leu^{587} bond in the α_2 chain of laminin 5 by MMP-2 induces migration of normal breast epithelial cells by exposing a cryptic promigratory site on laminin 5 (76). The cleaved form of laminin 5 was found in tumors and in tissues undergoing remodeling but not in quiescent tissues (76). Amnion epithelial cells undergo apoptotic cell death before the onset of labor, which is closely associated with degradation of type I collagen by collagenase (most likely MMP-13) (77). Cleavage of type I collagen by MMP-1 and by MMP-13 initiates keratinocyte migration during reepithelialization (78) and osteoclast activation (79), respectively.

Further insights into biological and pathological functions of matrixins have been provided by the use of transgenic animals and gene transfer techniques (80). The expression of MMP-9 in trophoblast giant cells appears to be critical during mouse blastocyst outgrowth and in early implantation stages *in vivo* (81, 82) and in later embryonic skeletal tissue development (83). Vu *et al.* (84) reported that MMP-9-deficient ($MMP\text{-}9^{-/-}$) mice exhibited no obvious phenotypic defects. However, these mice exhibited a delayed long bone growth associated with an abnormally thickened growth plate, which was accompanied by delayed apoptosis of hypertrophic chondrocytes, vascularization, and ossification. This abnormality was detected only during development, and normal appearance of bone was seen by 8 weeks of age; probably the deficiency was compensated by other MMPs. $MMP\text{-}9^{-/-}$ mice are resistant to subepidermal blistering in experimental bullous pemphigoid, an autoimmune blistering disease, due to the lack of MMP-9 in neutrophils (85).

Mice deficient in MMP-12 (metalloelastase), in contrast to the wild-type, did not develop emphysema in response to long term exposure to cigarette smoke (86). $MMP\text{-}12^{-/-}$ macrophages are unable to penetrate basement membrane *in vitro* and *in vivo* (87).

The matrixins are considered to participate in dissemination of cancer cells by breaking down ECM, but they are also important in creating an environment that supports the initiation and maintenance of growth of primary and metastatic tumors (88). Mice with adenomatous polyposis coli (*Apc*) gene mutation develop spontaneous intestinal tumors, but such mice deficient in MMP-7 showed significantly reduced tumor multiplicity and growth (89). MMP-11-deficient mice also showed reduced chemically induced tumorigenesis (90). Those effects may be related to proteolytic release of growth factors and/or alteration of cellular environments.

Future Prospects

With a wealth of information provided in this field over the last few years we have begun to understand the significance of matrixins in biology and pathology. The actions of matrixins *in vivo* are complex and diverse; they are not restricted to simple breakdown of ECM, but they may reveal cryptic biological functions of ECM macromolecules. Both concepts need to be taken into consideration for our understanding of the timely alteration of cellular environments required in normal development and morphogenesis. We also anticipate the discovery of many more new MMPs, which will introduce further complexity in tissue matrix catabolism. Such discovery will, however, help us reveal more precise mechanisms of tissue matrix turnover. Detailed structural and functional analyses of MMPs led to the development of numerous potent synthetic inhibitors of matrixins, and some are in clinical trials to treat patients with cancer, arthritis, periodontal disease, and corneal ulceration. Such agents may be of great therapeutic value, but concerns remain about the consequences of inhibiting biologically functioning matrixins and related ADAMs. Alternative approaches may be tissue-targeted gene therapy with TIMPs or TIMP variants that selectively inhibit specific metalloproteinases. The further gain of knowledge about the mechanisms of cell/tissue-specific regulation of MMP gene expression and their signal transduction pathways may also lead to the rational design of inhibitors that perturb the production of MMPs in a particular cell type without affecting other cells. Such agents will be of great value not only for understanding of the basic biology of matrix and its turnover but also for intervention of diseases resulting from aberrant ECM degradation.

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REFERENCES

1. Nagase, H. (1996) in *Zinc Metalloproteases in Health and Disease* (Hooper, N. M., ed) pp. 153–204, Taylor & Francis, London
2. Fini, M. E., Cook, J. R., Mohan, R., and Brinckerhoft, C. E. (1998) in *Matrix Metalloproteinases* (Parks, W. C., and Mecham, R. P., eds) pp. 299–356, Academic Press, San Diego
3. Lepage, T., and Gache, C. (1990) *EMBO J.* **9,** 3003–3012

4. Wada, K., Sato, H., Kinoh, H., Kajita, M., Yamamoto, H., and Seiki, M. (1998) *Gene* (*Amst.*) **211,** 57–62
5. Pak, J. H., Liu, C. Y., Huangpu, J., and Graham, J. S. (1997) *FEBS Lett.* **404,** 283–288
6. Massova, I., Kotra, L. P., Fridman, R., and Mobashery, S. (1998) *FASEB J.* **12,** 1075–1095
7. Barrett, A. J., Rawlings, N. D., and Woessner, J. F. (1998) *Handbook of Proteolytic Enzymes*, Academic Press, London
8. Van Wart, H. E., and Birkedal-Hansen, H. (1990) *Proc. Natl. Acad. Sci. U. S. A.* **87,** 5578–5582
9. Becker, J. W., Marcy, A. I., Rokosz, L. L., Axel, M. G., Burbaum, J. J., Fitzgerald, P. M., Cameron, P. M., Esser, C. K., Hagmann, W. K., Hermes, J. D., and Springer, J. P. (1995) *Protein Sci.* **4,** 1966–1976
10. Gururajan, R., Grenet, J., Lahti, J. M., and Kidd, V. J. (1998) *Genomics* **52,** 101–106
11. Pei, D., and Weiss, S. J. (1995) *Nature* **375,** 244–247
12. Pei, D., and Weiss, S. J. (1996) *J. Biol. Chem.* **271,** 9135–9140
13. Bode, W., Gomis-Rüth, F. X., and Stöcker, W. (1993) *FEBS Lett.* **331,** 134–140
14. Dhanaraj, V., Ye, Q. Z., Johnson, L. L., Hupe, D. J., Ortwine, D. F., Dunbar, J. B., Jr., Rubin, J. R., Pavlovsky, A., Humblet, C., and Blundell, T. L. (1996) *Structure* **4,** 375–386
15. Allan, J. A., Docherty, A. J. P., Barker, P. J., Huskisson, N. S., Reynolds, J. J., and Murphy, G. (1995) *Biochem. J.* **309,** 299–306
16. Steffensen, B., Wallon, U. M., and Overall, C. M. (1995) *J. Biol. Chem.* **270,** 11555–11566
17. Gomis-Rüth, F. X., Gohlke, U., Betz, M., Knäuper, V., Murphy, G., López-Otín, C., and Bode, W. (1996) *J. Mol. Biol.* **264,** 556–566
18. Bode, W. (1995) *Structure* **3,** 527–530
19. Clark, I. M., and Cawston, T. E. (1989) *Biochem. J.* **263,** 201–206
20. Murphy, G., Willenbrock, F., Ward, R. V., Cockett, M. I., Eaton, D., and Docherty, A. J. P. (1992) *Biochem. J.* **283,** 637–641
21. Strongin, A. Y., Collier, I., Bannikov, G., Marmer, B. L., Grant, G. A., and Goldberg, G. I. (1995) *J. Biol. Chem.* **270,** 5331–5338
22. de Souza, S. J., Pereira, H. M., Jacchieri, S., and Brentani, R. R. (1996) *FASEB J.* **10,** 927–930
23. Guo, H., Zucker, S., Gordon, M. K., Toole, B. P., and Biswas, C. (1997) *J. Biol. Chem.* **272,** 24–27
24. Aoudjit, F., Potworowski, E. F., and St-Pierre, Y. (1998) *J. Immunol.* **160,** 2967–2973
25. Romanic, A. M., and Madri, J. A. (1994) *J. Cell Biol.* **125,** 1165–1178
26. Malik, N., Greenfield, B. W., Wahl, A. F., and Kiener, P. A. (1996) *J. Immunol.* **156,** 3952–3960
27. Xie, B., Laouar, A., and Huberman, E. (1998) *J. Biol. Chem.* **273,** 11576–11582
28. Kadono, Y., Okada, Y., Namiki, M., Seiki, M., and Sato, H. (1998) *Cancer Res.* **58,** 2240–2244
29. Haas, T. L., Davis, S. J., and Madri, J. A. (1998) *J. Biol. Chem.* **273,** 3604–3610
30. Seltzer, J. L., Lee, A. Y., Akers, K. T., Sudbeck, B., Southon, E. A., Wayner, E. A., and Eisen, A. Z. (1994) *Exp. Cell Res.* **213,** 365–374
31. Gilles, C., Polette, M., Seiki, M., Birembaut, P., and Thompson, E. W. (1997) *Lab. Invest.* **76,** 651–660
32. Werb, Z., Tremble, P. M., Behrendtsen, O., Crowley, E., and Damsky, C. H. (1989) *J. Cell Biol.* **109,** 877–889
33. Kheradmand, F., Werner, E., Tremble, P., Symons, M., and Werb, Z. (1998) *Science* **280,** 898–902
34. Tremble, P., Damsky, C. H., and Werb, Z. (1995) *J. Cell Biol.* **129,** 1707–1720
35. Spiegel, S., Foster, D., and Kolesnick, R. (1996) *Curr. Opin. Cell Biol.* **8,** 159–167
36. Reunanen, N., Westermarck, J., Häkkinen, L., Holmström, T. H., Elo, I., Eriksson, J. E., and Kähäri, V. M. (1998) *J. Biol. Chem.* **273,** 5137–5145
37. Korzus, E., Nagase, H., Rydell, R., and Travis, J. (1997) *J. Biol. Chem.* **272,** 1188–1196
38. Fisher, G. J., Datta, S. C., Talwar, H. S., Wang, Z. Q., Varani, J., Kang, S., and Voorhees, J. J. (1996) *Nature* **379,** 335–339
39. Brenneisen, P., Wenk, J., Klotz, L. O., Wlaschek, M., Briviba, K., Krieg, T., Sies, H., and Scharffetter-Kochanek, K. (1998) *J. Biol. Chem.* **273,** 5279–5287
40. Fisher, G. J., Talwar, H. S., Lin, J. Y., Lin, P. P., McPhillips, F., Wang, Z. Q., Li, X. Y., Wan, Y. S., Kang, S. W., and Voorhees, J. J. (1998) *J. Clin. Invest.* **101,** 1432–1440
41. Nagase, H. (1997) *Biol. Chem.* **378,** 151–160
42. Carmeliet, P., Moons, L., Lijnen, R., Baes, M., Lemaitre, V., Tipping, P., Drew, A., Eeckhout, Y., Shapiro, S., Lupu, F., and Collen, D. (1997) *Nat. Genet.* **17,** 439–444
43. Sato, H., Takino, T., Okada, Y., Cao, J., Shinagawa, A., Yamamoto, E., and Seiki, M. (1994) *Nature* **370,** 61–65
44. Butler, G. S., Butler, M. J., Atkinson, S. J., Will, H., Tamura, T., Vanwestrum, S. S., Crabbe, T., Clements, J., d'Ortho, M. P., and Murphy, G. (1998) *J. Biol. Chem.* **273,** 871–880
45. Kinoshita, T., Sato, H., Okada, A., Ohuchi, E., Imai, K., Okada, Y., and Seiki, M. (1998) *J. Biol. Chem.* **273,** 16098–16103
46. Mazzieri, R., Masiero, L., Zanetta, L., Monea, S., Onisto, M., Garbisa, S., and Mignatti, P. (1997) *EMBO J.* **16,** 2319–2332
47. Nakahara, H., Howard, L., Thompson, E. W., Sato, H., Seiki, M., Yeh, Y. Y., and Chen, W. T. (1997) *Proc. Natl. Acad. Sci. U. S. A.* **94,** 7959–7964
48. Hiraoka, N., Allen, E., Apel, I. J., Gyetko, M. R., and Weiss, S. J. (1998) *Cell* **95,** 365–377
49. Brooks, P. C., Stromblad, S., Sanders, L. C., von Schalscha, T. L., Aimes, R. T., Stetler-Stevenson, W. G., Quigley, J. P., and Cheresh, D. A. (1996) *Cell* **85,** 683–693
50. Brooks, P. C., Silletti, S., von Schalscha, T. L., Friedlander, M., and Cheresh, D. A. (1998) *Cell* **92,** 391–400
51. Bafetti, L. M., Young, T. N., Itoh, Y., and Stack, M. S. (1998) *J. Biol. Chem.* **273,** 143–149
52. Gomez, D. E., Alonso, D. F., Yoshiji, H., and Thorgeirsson, U. P. (1997) *Eur. J. Cell Biol.* **74,** 111–122
53. Gomis-Rüth, F. X., Maskos, K., Betz, M., Bergner, A., Huber, R., Suzuki, K., Yoshida, N., Nagase, H., Brew, K., Bourenkov, G. P., Bartunik, H., and Bode, W. (1997) *Nature* **389,** 77–81
54. Fernandez-Catalan, C., Bode, W., Huber, R., Turk, D., Calvete, J. J., Lichte, A., Tschesche, H., and Maskos, K. (1998) *EMBO J.* **17,** 5238–5248
55. Muskett, F. W., Frenkiel, T. A., Feeney, J., Freedman, R. B., Carr, M. D., and Williamson, R. A. (1998) *J. Biol. Chem.* **273,** 21736–21743
56. Tuuttila, A., Morgunova, E., Bergmann, U., Lindqvist, Y., Maskos, K., Fernandez-Catalan, C., Bode, W., Tryggvason, K., and Schneider, G. (1998) *J. Mol. Biol.* **284,** 1133–1140
57. Arumugam, S., Hemme, C. L., Yoshida, N., Suzuki, K., Nagase, H., Bejanskii, M., Wu, B., and Van Doren, S. R. (1998) *Biochemistry* **37,** 9650–9657
58. Murphy, A. N., Unsworth, E. J., and Stetler-Stevenson, W. G. (1993) *J. Cell. Physiol.* **157,** 351–358
59. Hayakawa, T., Yamashita, K., Ohuchi, E., and Shinagawa, A. (1994) *J. Cell Sci.* **107,** 2373–2379
60. Chesler, L., Golde, D. W., Bersch, N., and Johnson, M. D. (1995) *Blood* **86,** 4506–4515
61. Boujrad, N., Ogwuegbu, S. O., Garnier, M., Lee, C. H., Martin, B. M., and Papadopoulos, V. (1995) *Science* **268,** 1609–1612
62. Nothnick, W. B., Soloway, P., and Curry, T. E. (1997) *Biol. Reprod.* **56,** 1181–1188
63. Clark, I. M., Powell, L. K., and Cawston, T. E. (1994) *Biochem. Biophys. Res. Commun.* **203,** 874–880
64. Zhao, W. Q., Li, H., Yamashita, K., Guo, X. K., Hoshino, T., Yoshida, S., Shinya, T., and Hayakawa, T. (1998) *J. Cell Sci.* **111,** 1147–1153
65. Smith, M. R., Kung, H. F., Durum, S. K., Colburn, N. H., and Sun, Y. (1997) *Cytokine* **9,** 770–780
66. Ahonen, M., Baker, A. H., and Kähäri, V. M. (1998) *Cancer Res.* **58,** 2310–2315
67. Amour, A., Slocombe, P. M., Webster, A., Butler, M., Knight, C. G., Smith, B. J., Stephens, P. E., Shelley, C., Hutton, M., Knäuper, V., Docherty, A. J. P., and Murphy, G. (1998) *FEBS Lett.* **435,** 39–44
68. Hargreaves, P. G., Wang, F. F., Antcliff, J., Murphy, G., Lawry, J., Russell, R. G. G., and Croucher, P. I. (1998) *Br. J. Haematol.* **101,** 694–702
69. Baker, A. H., Zaltsman, A. B., George, S. J., and Newby, A. C. (1998) *J. Clin. Invest.* **101,** 1478–1487
70. Guedez, L., Stetler-Stevenson, W. G., Wolff, L., Wang, J., Fukushima, P., Mansoor, A., and Stetler-Stevenson, M. (1998) *J. Clin. Invest.* **102,** 2002–2010
71. Pavloff, N., Staskus, P. W., Kishnani, N. S., and Hawkes, S. P. (1992) *J. Biol. Chem.* **267,** 17321–17326
72. Langton, K. P., Barker, M. D., and McKie, N. (1998) *J. Biol. Chem.* **273,** 16778–16781
73. Felbor, U., and Weber, B. H. (1998) *Ophthalmologe* **95,** 287–290
74. Parks, W. C., and Mecham, R. P. (1998) *Matrix Metalloproteinases*, Academic Press, San Diego
75. Zuo, J., Ferguson, T. A., Hernandez, Y. J., Stetler-Stevenson, W. G., and Muir, D. (1998) *J. Neurosci.* **18,** 5203–5211
76. Giannelli, G., Falkmarzillier, J., Schiraldi, O., Stetler-Stevenson, W. G., and Quaranta, V. (1997) *Science* **277,** 225–228
77. Lei, H., Furth, E. E., Kalluri, R., Chiou, T., Tilly, K. I., Tilly, J. L., Elkon, K. B., Jeffrey, J. J., and Strauss, J. F., III (1996) *J. Clin. Invest.* **98,** 1971–1978
78. Pilcher, B. K., Dumin, J. A., Sudbeck, B. D., Krane, S. M., Welgus, H. G., and Parks, W. C. (1997) *J. Cell Biol.* **137,** 1445–1457
79. Holliday, L. S., Welgus, H. G., Fliszar, C. J., Veith, G. M., Jeffrey, J. J., and Gluck, S. L. (1997) *J. Biol. Chem.* **272,** 22053–22058
80. Shapiro, S. D. (1998) *Curr. Opin. Cell Biol.* **10,** 602–608
81. Alexander, C. M., Hansell, E. J., Behrendtsen, O., Flannery, M. L., Kishnani, N. S., Hawkes, S. P., and Werb, Z. (1996) *Development* **122,** 1723–1736
82. Das, S. K., Yano, S., Wang, J., Edwards, D. R., Nagase, H., and Dey, S. K. (1997) *Dev. Genet.* **21,** 44–54
83. Reponen, P., Sahlberg, C., Munaut, C., Thesleff, I., and Tryggvason, K. (1994) *J. Cell Biol.* **124,** 1091–1102
84. Vu, T. H., Shipley, J. M., Bergers, G., Berger, J. E., Helms, J. A., Hanahan, D., Shapiro, S. D., Senior, R. M., and Werb, Z. (1998) *Cell* **93,** 411–422
85. Liu, Z., Shipley, J. M., Vu, T. H., Zhou, X. Y., Diaz, L. A., Werb, Z., and Senior, R. M. (1998) *J. Exp. Med.* **188,** 475–482
86. Hautamaki, R. D., Kobayashi, D. K., Senior, R. M., and Shapiro, S. D. (1997) *Science* **277,** 2002–2004
87. Shipley, J. M., Wesselschmidt, R. L., Kobayashi, D. K., Ley, T. J., and Shapiro, S. D. (1996) *Proc. Natl. Acad. Sci. U. S. A.* **93,** 3942–3946
88. Chambers, A. F., and Matrisian, L. M. (1997) *J. Natl. Cancer Inst.* **89,** 1260–1270
89. Wilson, C. L., Heppner, K. J., Labosky, P. A., Hogan, B. L., and Matrisian, L. M. (1997) *Proc. Natl. Acad. Sci. U. S. A.* **94,** 1402–1407
90. Masson, R., Lefebvre, O., Noel, A., El Fahime, M., Chenard, M. P., Wendling, C., Kebers, F., LeMeur, M., Dierich, A., Foidart, J.-M., Basset, P., and Rio, M.-C. (1998) *J. Cell Biol.* **140,** 1535–1541
91. Carson, M. (1997) *Methods Enzymol.* **277,** 493–505